

Machine Learning Project

[Background and Purpose](#)

[Anticipated Results and Deliverables](#)

[Project Details and Specifics](#)

[Annex: Examples](#)

[Annex: Tags on ReliefWeb](#)

Background and Purpose

ReliefWeb collects over 1,000 news sources in Inoreader, our primary content aggregation platform. Our editors then pick the articles according to our editorial guidelines and post them on www.reliefweb.int. The posting process involves a range of tagging and summarizing to make finding specific reports and information on ReliefWeb easier for the humanitarian community.

ReliefWeb wants to explore how Machine Learning could support our operations in the following ways:

1. Support the tagging of ReliefWeb documents to enhance overall consistency, aiming on creating an standardized service to be used by the humanitarian community to tag their reports.
2. Alleviate pressure and save time - ReliefWeb editors are stressed because of the amount of content coming in via Inoreader
3. Save monetary resources in the long run by being able to support a bigger operation with less editorial staff

Previous attempts and challenges

ReliefWeb has already tested the concept with two proofs of concept in 2018 and 2020. Although the results of both were promising, ReliefWeb lacked the resources to keep developing the project and operationalizing the models.

The first project tried to tag most of the fields for the reports using diverse python libraries (not only Machine Learning libraries). The second project only tried to tag the field career category of the jobs.

The first project lacked a stable model that could be replicable and it didn't measure the accuracy of the results. It demonstrated that the tagging problem can be solved using different libraries.

The second project focused more on the model and got an accuracy of 71% (versus 84% human accuracy). Some of the challenges in this project were: the accuracy was not enough, to avoid bias for over-represented values, support texts in different languages, be able to read from PDF documents, the algorithm should be able to learn again from corrections, operationalizing this model for other fields and in a real production environment.

Anticipated Results and Deliverables

Depending on feasibility and complexity of our tagging system, we are hoping that all tagging could be done automatically, directly from RSS feeds or by URLs entered by the editors.

Editors may then need to proof (review) document posting. This would save a lot of time as proofing documents is less time consuming than tagging.

As ReliefWeb's technical and financial resources are limited, we would expect a fully developed algorithm to automatically tag reports. ReliefWeb could make some editors available to test the algorithm and provide feedback.

Project Details and Specifics

Inoreader collects our documents in a range of different formats, mostly linking to websites, but also linking to PDF documents. It also collects documents in English, Spanish and French (majority English), and we sometimes receive documents in Portuguese, Ukrainian, or other languages. Editors currently read documents and then manually upload and tag the documents according to the tags below.

Annex: Examples

- Example of a PDF document collected via Inoreader:

Situation report #3, December 2019

2020-06-11(Thu) 10:09 via WHO/Europe - Sitreps: Measles in Europe ▾



1-Core

Add tags

Create note

Like Broadcast



Add your comment...

The document title is a URL directing readers to the [WHO Europe website](#), where the actual [PDF document](#) can be found. The tagging information would have to be taken from the PDF document, not the WHO Europe website.

- Another example of a PDF report collected via Inoreader:

UN releases US\$65 million for humanitarian response in Ethiopia's Tigray crisis worsens

2021-05-06(Thu) 12:23 via CERF - Press Releases ▾



(New York, 6 May 2021): The United Nations has released US\$65 million for the humanitarian response in Ethiopia. More than 16 million people need humanitarian assistance throughout Ethiopia, including an estimated 4.5 million in the Tigray region.

The \$65 million...

1-Core

Core1

Add tags

Create note

Like Broadcast



Add your comment...

The document title is a URL directing readers directly to the [PDF](#) where all information relevant for posting and tagging can be found.

- An example of a report directly embedded on a website collected via Inoreader:

Haiti earthquake: EU continues to mobilise emergency assistance

85 2021-08-19(Thu) 06:38 by Hans via ECHO - News



In addition to the €3 million in humanitarian funding mobilised by EU to address the most urgent needs in the immediate aftermath of the earthquake, the EU is scaling up its assistance.

1-Core Core3 Add tags

Create note

Like Broadcast



Add your comment...

The title URL directs readers to the [ECHO website](#), where all information relevant for posting and tagging can be found.

Annex: Tags on ReliefWeb

ReliefWeb tags should be filled in according to the text of the report. They include the following:

1. Language - checkbox
2. Title - text field
3. Body - text field (usually for a short summary of the report, or if report is a press release, entire press release goes here)
4. Files
 - Attachments - PDF is attached here (multiple if available in multiple languages)

→ Image - Image is attached here, if any (also depends on copyrights)

5. Dates

→ Publication date - date field (date report was originally published)

→ Embargo date - date field (if report is embargoed, it will automatically be published on that date - in UTC/GMT)

6. Source

→ Organization - multiple select field

→ Origin - checkbox (URL, Submit mailbox, Reliefweb product)

→ Origin notes - text field (for origin URL or if origin checkbox is Submit mailbox, email addresses of sender)

7. Countries and disasters

→ Country - multiple select field

→ Primary country - single select field

→ Disaster - multiple select field

→ Disaster type - automatically selected based on disaster above

8. Details

→ Content format - single select field

→ Theme - multiple select field

9. Headline information - *skip for now*

10. Editorial flags - *skip for now*

11. Revisions - *skip not needed* - this field provides space for comments when editors proof-read documents